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A DETAILED PALEOMAGNETIC SURVEY OF THE TYPE LOCATION FOR THE GILSA GEOMAGNETIC POLARITY EVENT

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A total of 163 cores have been taken from a maximum of 40 separate lavas in three separate sections of the Jökuldalur, southwest of Egilsstadir, Iceland, and subjected to paleomagnetic analysis and some K–Ar dating. Previous work on the sections by McDougall and Wensink (1966) led to the establishment of the Gilsá geomagnetic polarity event, with an age of about 1.60 m.y., during the reversed polarity Matuyama epoch. This earlier study described a possible reversely magnetized lava separating the Gilsá event from a second normal polarity lava, perhaps representing the Olduvai event. Such a possibility was subsequently a source of speculation from diverse sources concerning the polarity history for the lower Matuyama. The present study indicates clearly that there is no second normal polarity event represented in the sections. Only one normal polarity event is therefore represented in the lower Matuyama of the Jökuldalur, and the age of the lavas involved is confirmed to be approximately 1.58 ± 0.08 m.y. Because of uncertainties in the interpretation of the original K–Ar results from Olduvai Gorge, it is still not possible to be certain that the Olduvai and Gilsá events are separated in time. An incidental result of the survey is evidence to show that, contrary to recent suggestions by Einarsson (1972), there is no substantial hiatus between the major lower parts of the section and lavas believed to represent extrusions after a regional tilting and peneplanation.

1. Introduction

As summarized by Watkins [1] the Olduvai event was the first period of normal magnetic polarity identified during the Matuyama reversed polarity epoch [2]. It was believed for some years that this event occurred at about t = 1.9 m.y. Therefore, when McDougall and Wensink [3] reported in 1966, on the basis of K–Ar and paleomagnetic data from Jökuldalur in east-central Iceland (Fig. 1), a normal polarity lava of 1.60 m.y. age, it appeared that a second normal polarity event had been discovered. Such a possibility was subsequently enhanced by the report [3] of a *possibly* reversely magnetized lava overlying a second normal polarity lava (lava no. 16 in Fig. 2) in the section examined (Fig. 3). This second normal polarity lava has been frequently referred to as supporting evidence for complexities in lower Matuyama geomagnetic polarity history, recorded by mid-oceanic magnetic anomaly patterns [4–6], and deep-sea sedimentary cores [7–10]. Because of its global significance, therefore, it has become clearly appropriate to provide unambiguous resolution of the existence and age of the suspected second normal polarity event. In this communication we describe the results of an intensive paleomagnetic and K–Ar survey of the sections

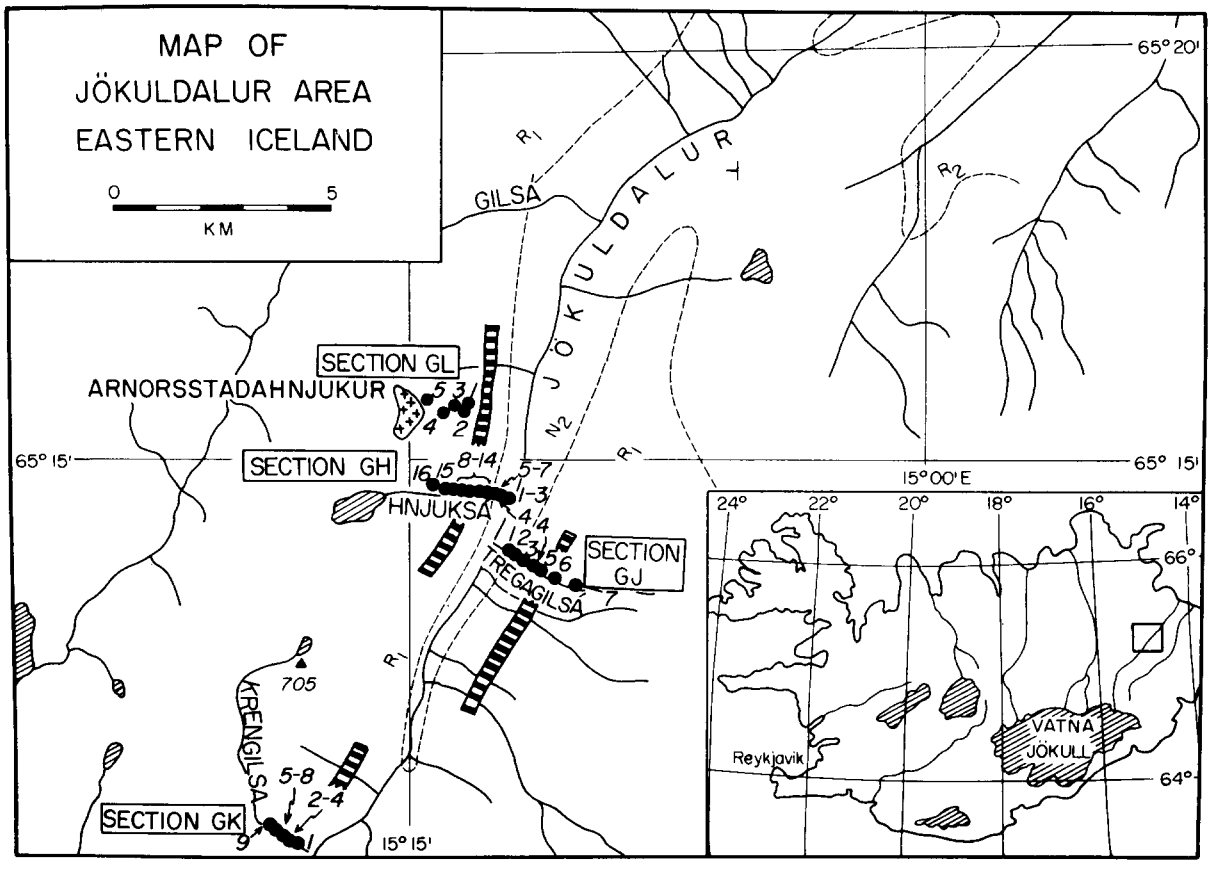


Fig. 1. Map showing location of the three sampled sections. Base map taken from Wensink [12]. Site numbers correspond to those units given in Fig. 2. For the inferred correlation between the original sites in the Hnjuksá stream [12,13] and the present survey, see Fig. 2. N_2 and R_1 polarity boundary is according to Wensink [12]. This has now been identified as the Gauss-Matuyama polarity boundary at $t = 2.41$ m.y. Heavy stripes = "Magnetically normal flows in R_1 series", according to Wensink [13]. Diagonal stripes = lake. Crosses = tuff-breccia.

where what is now called the Gilsá event was originally detected. At the same time, the survey is relevant to regional geological history most recently discussed by Einarsson [11], who has questioned the limited range of the earlier K–Ar dates [3] since he believes that evidence exists to support a model involving regional tilting and subsequent peneplanation, indicative of substantial time being represented.

2. Methods

2.1. Field

The critical sections were documented in unusual detail by Wensink [12,13]. In Fig. 2 we include Wensink's original stratigraphic compilation and his lava identification numbers. The locations of the three sections which we have surveyed in detail are shown in Fig. 1. These three sections were selected because of the specific indication by Wensink [12,13] that a normal polarity lava was present in the "R_1" part of all three sections as shown in Fig. 1. In particular, the location of lavas 12 through 17 as given by Wensink [12] in the Hnjuksá stream section of Fig. 1, includes the critical possibly reversely magnetized lava between units 15 and 17 (Figs. 2 and 3), showing that this section alone should provide the means to resolve the ambiguity under investigation. The only doubt that this would be the case is provided by table 1 of Wensink [13], in which the paleomagnetic data

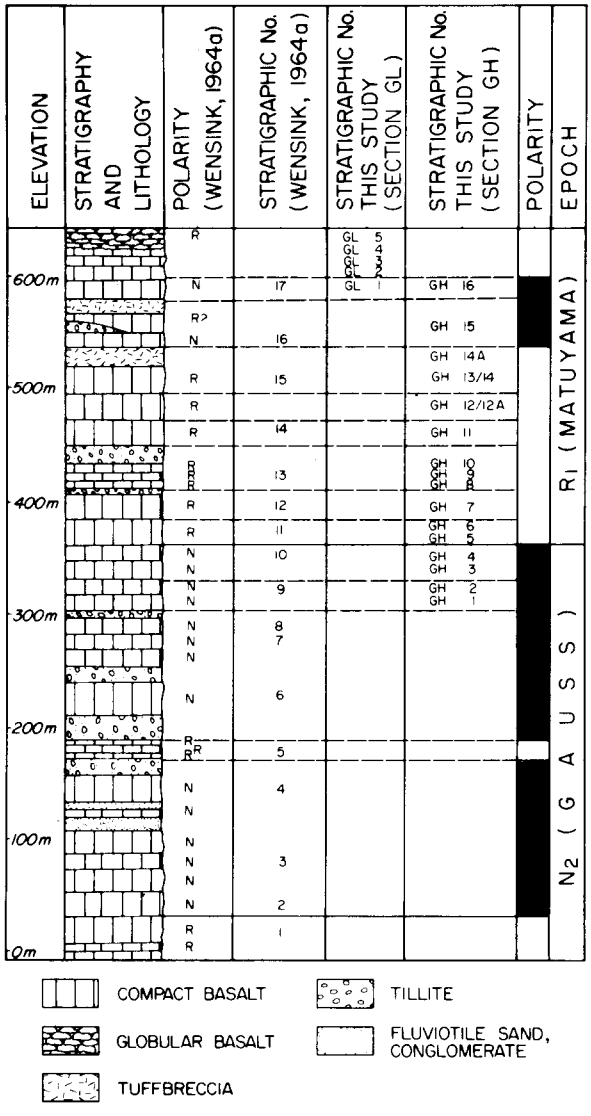


Fig. 2. Stratigraphy of the Jökuldalur area, together with original and present magnetostratigraphic results. From left to right: elevation in meters above sea level; stratigraphy and lithology [12]; polarity measured in original survey [12] R = reversed, N = normal polarity; stratigraphic number in original survey [3,12,13]; stratigraphic number in present survey; measured polarity in present survey (see Table 1); polarity epoch (N_2 and R_1 from original survey; Gauss and Matuyama from subsequent development of terminology).

for his units 1 through 17 (Fig. 2) are presented, but omitted are the results for unit 16 in particular, as well as the possibly reversely magnetized lava (to

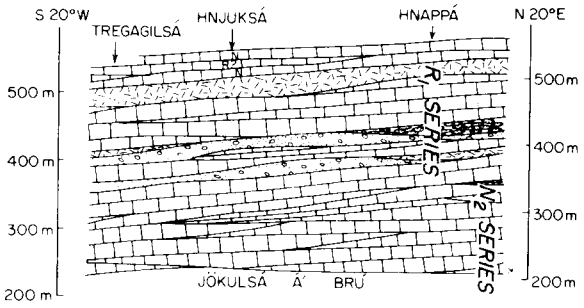


Fig. 3. Sketch of magnetostratigraphy of Hnjuksá and Tregagilsá stream sections (Fig. 1), from Wensink [13]. The "possibly" reversely magnetized lava is shown at the top of the sequence, separating an otherwise single normal polarity sequence into two distinct units.

which no number was assigned). In 1971 and 1973, therefore, we conducted an intensive paleomagnetic survey of the Hnjuksá, Tregagilsá, and Krengilsá stream sections (Fig. 1).

All possible separate units were sampled, using a portable gasoline-powered drill. At least four separately oriented cores of 2.5 cm diameter were taken from each body. Exposures are not complete in the upper part of the Hnjuksá stream, but because of the particular importance of the sequence, we sampled outcrops which were not definitely distinct units. Thus units 12A, 13, 14A, and 15A of Table 1a could have represented repeated sampling of other bodies. A total of 19 separate outcrops were sampled (GH1 to 16 in Fig. 2 and Table 1a) not only to include units 9 through 17 of Wensink [12], as included in Fig. 2, but also an upwards extension by a search to the west (Fig. 1), and by sampling the Arnorsstadahnjukur "Tuff-breccia Volcano" which lies above the "peneplained" surface of the valley (Fig. 1), according to Einarsson [11]. Units GL1 to 5 are from this latter section. GL1 is possibly a repeat of GH16 (Fig. 3). Hand samples were taken from the least weathered parts of all units, for petrographic and possible K–Ar analyses.

2.2. *Laboratory*

All cores were sliced into specimens of 2.2 cm length. Remanent magnetism direction and intensities were measured on spinner magnetometers. Following tests on randomly selected specimens, unstable com-

TABLE 1

Paleomagnetic data for the Jökuldalur region, east-central Iceland

No.	N	D (deg.)	I (deg.)	R	K	ϕ'	θ'
(a) Sections GH and GL (Hnjuksá and Arnorsstadahnjukur)							
GL5	3	307.7	−72.8	2.9586	48.3	197.5	−39.5
GL4	4	323.4	−80.3	3.9768	129.1	181.7	−48.5
GL3	4	222.0	−63.4	3.9941	507.0	275.6	−59.6
GL2	4	179.4	−58.2	3.9836	182.6	345.9	−63.7
GL1	4	327.3	73.2	3.9979	1397.3	247.4	73.6
GH16	4	341.0	78.4	3.9774	132.7	281.9	82.1
GH15A	4	158.9	85.7	3.9896	287.1	350.4	57.0
GH15	8	343.7	83.1	7.5855	16.8	327.0	77.5
GH14A	4	199.8	−63.7	3.9895	1857.0	306.6	−67.3
GH14	4	155.8	−76.9	3.9984	286.2	84.7	−79.9
GH13	4	156.0	−76.8	3.9976	1230.7	84.3	−79.9
GH12A	4	247.4	−87.4	3.9935	465.1	177.1	−66.7
GH12	4	172.5	−84.1	3.9758	124.1	158.1	−76.7
GH11	4	258.6	−80.9	3.9873	235.7	205.7	−62.8
GH10	4	206.7	−65.6	3.9983	1800.6	292.6	−67.5
GH9	4	222.9	−68.2	3.9956	689.1	266.5	−64.3
GH8	4	213.6	−76.6	3.9951	612.6	242.0	−75.9
GH7	4	176.0	−79.1	3.9946	555.3	143.7	−86.0
GH6	4	203.3	−78.6	3.9864	220.4	227.2	−80.4
GH5	4	86.8	−59.6	3.9677	93.0	97.0	−34.8
GH4	4	40.3	75.3	3.9918	366.1	65.3	72.2
GH3	4	37.3	74.6	3.9971	1032.1	71.2	72.9
GH2	4	300.9	75.5	3.9957	696.1	276.7	64.8
GH1	4	14.4	77.1	3.9977	1332.9	66.5	84.0
(b) Section GK (Krengilsá)							
GK9	4	155.7	−55.3	3.9733	112.4	22.8	−57.3
GK8	4	183.1	−63.2	3.9965	849.3	338.6	−69.4
GK7	4	136.5	−72.3	3.9849	198.6	74.5	−68.2
GK6	4	211.1	−70.0	3.9926	405.6	276.0	−71.0
GK5	4	86.6	86.0	3.9959	739.5	3.5	64.5
GK4	4	39.8	83.8	3.9889	271.5	12.3	72.9
GK3	4	13.7	69.2	3.9680	93.9	129.0	75.8
GK2	4	36.1	69.9	3.9945	540.7	89.9	68.8
GK1	4	287.8	−48.0	2.3804*	1.9	226.6	19.1
(c) Section GJ (Tregagilsá)							
GJ7	4	174.5	−74.2	3.9942	520.6	15.1	−84.6
GJ6	4	155.3	−86.2	3.9837	184.1	154.7	−71.8
GJ5	4	113.3	−85.8	3.9953	641.3	144.4	−67.3
GJ4	4	133.1	−78.9	3.9947	568.4	106.7	−71.8
GJ3	4	199.2	−85.6	3.9895	285.7	174.9	−73.3
GJ2	4	295.1	75.9	3.9953	634.0	281.2	63.0
GJ1	4	275.2	75.1	3.9945	545.8	290.1	55.0

For section location see Fig. 1. No. = numbers of lava in section (Fig. 2). Bodies are given in stratigraphic order, with lowest number = lowest lava. N = number of separately oriented cores. D and I = declination (in degrees east of north) and inclination (with respect to horizontal; negative = upwards) of mean direction of remanent magnetism, after demagnetization in a peak alternating magnetic field of 100 or 200 Oe, selection being made to obtain minimum scatter of directions. R = resultant vector, applying unit vector per core. $K = (N - 1)/(N - R)$. θ' and ϕ' = latitude and longitude of virtual geomagnetic pole corresponding to D and I. θ' in degrees north or south (negative values); ϕ' in degrees east of Greenwich. * = statistically random at 95% confidence level. All data corrected for slight regional dip.

ponents in all specimens were minimized by the application of peak alternating magnetic fields of 100 and 200 Oe. Mean directions of remanent magnetism for each lava were then computed using a minimum dispersion criterion [14].

A thin section, made from each sample collected, was examined using a petrographic microscope, in order to restrict dating to alteration-free materials.

Samples selected for dating were crushed to a fragment size of 1.6–0.6 mm from which an aliquot was taken and reduced to less than 0.15 mm for potassium analysis, carried out by a flame photometric technique as described previously [15], using a modified Baird Atomic KY3 flame photometer. The coarser material was used for argon extractions: 15–30 g of rock was fused in each case in a high-vacuum extraction line and the purified argon isotopically analyzed in a modified AEI MS10 mass spectrometer, which has a 4.2-kilogauss permanent magnet and a digital readout system [16]. The method of isotope dilution was employed, using ^{38}Ar prepared from a gas pipette as the tracer.

3. Results and discussion

3.1. Paleomagnetic properties

Paleomagnetic data for each lava in the three different sections are given in Table 1. Fisher statistical parameters [17] are included for each lava. The data are presented in stratigraphic order in each section.

The intensities of magnetization of all cores are in the range 10^{-3} to 10^{-4} emu g^{-1}. Median destructive fields are highly variable within each lava. Because magnetic stability in basalts depends primarily on the high-temperature titanomagnetite oxidation state, and titanomagnetite crystal size [18] it can be reliably inferred that the titanomagnetites are therefore of variable oxidation state. This provides evidence that the ambient geomagnetic field during the initial cooling (rather than any chemical factor) is the cause of the observed polarities, which do not vary within any lava. Only in lava GH15 is there any indication of consistent low stability: here (Table 3) some specimens lose over 90% of their intensity of natural remanent magnetism (J_0) when treated in alternating magnetic fields of only 100 Oe.

The virtual geomagnetic poles for each unit sampled in the Hnjuská stream (Table 1a) vary only slightly in most cases, but sufficiently to suggest that different bodies were in fact represented by most of the Hnjuksá units. Agreement with Wensink's [12] paleomagnetic data for each lava is good, providing evidence of satisfactory correlations between the two collections (Fig. 2).

3.2. Petrographic analyses

Most samples in the collection have abundant microvesicles. Samples chosen for K–Ar dating were virtually free of alteration, although all samples contain 2–5% of pale brown glass, which is fresh and isotropic to slightly anisotropic. Brief petrographic descriptions are given in the Appendix.

As the samples were known to have a low potassium content [3], it was considered probable that most of the potassium resides in the glass, so that the accuracy of any K–Ar ages obtained is likely to be primarily dependent on the argon retention properties of the glass.

3.3. Potassium-argon results

The K–Ar results are given in Table 2: errors are quoted at the level of one standard deviation as determined from the statistical uncertainties in each analysis, as previously described [19]. In a footnote to Table 2 we list the published ages of McDougall and Wensink [3] on two lavas from the same stratigraphic sequence.

The new age results are somewhat disappointing. Firstly, the replicate argon measurements are generally not in good agreement: in many cases the replicate calculated ages differ substantially more than expected from the statistical uncertainties. We are unable to present a satisfactory explanation for this behavior except to say that the evidence points to some form of inhomogeneity in the distribution of radiogenic argon and potassium. Normally, the relatively large sample weights used in the argon extractions and the small fragment size of the samples is sufficient to overcome problems of this kind. Secondly, although the measured ages are broadly consistent with the stratigraphy, there are discrepancies (Table 2). In particular, the results on sample GL2 clearly are aberrant as the

TABLE 2

K–Ar ages of basalts from the Jökuldalur region, east-central Iceland

Field No.	Lab No.	K (wt. %)	$^{40}Ar_{rad}$ (10^{-13} mol/g)	$^{40}Ar_{rad}/^{40}Ar_{total}$ × 100	Calculated age (m.y.) ± 1 s.d.	Mean age (m.y.) ± 1 s.d.	Polarity
GL4	73-1447	0.3280, 0.3295	7.49 8.04	14.9 19.6	1.28 ± 0.03 1.37 ± 0.03	1.32 ± 0.06	R
GL2	73-1449	0.2650, 0.2653	4.96 5.32	4.7 6.0	1.05 ± 0.07 1.13 ± 0.06	1.09 ± 0.06	R
GL1	73-1450	0.2359, 0.2348	6.17 6.34 6.71	4.0 3.9 3.8	1.47 ± 0.10 1.51 ± 0.11 1.60 ± 0.12	1.53 ± 0.07	N
GH16	73-1445	0.2450, 0.2463	6.78 6.84	19.9 24.7	1.55 ± 0.03 1.56 ± 0.02	1.55 ± 0.01	N
GH15-2	73-1444	0.1682, 0.1705	5.01 4.60 5.32 5.40	25.8 19.7 31.2 22.3	1.66 ± 0.04 1.53 ± 0.03 1.76 ± 0.04 1.79 ± 0.03	1.68 ± 0.12	N
GH15-1	73-1443	0.1452, 0.1464	4.22 4.23	22.7 21.8	1.62 ± 0.03 1.63 ± 0.03	1.63 ± 0.01	N
GH15	73-555	0.1533, 0.1534	4.01 4.55	11.2 26.7	1.47 ± 0.04 1.67 ± 0.03	1.57 ± 0.14	N
GH11	73-554	0.5122, 0.5131	17.89 18.60	33.8 45.1	1.96 ± 0.02 2.04 ± 0.02	2.00 ± 0.06	R

$\lambda_e = 0.585 \times 10^{-10}$ yr^{-1}; $\lambda_\beta = 4.72 \times 10^{-10}$ yr^{-1}; $^{40}K/K = 1.19 \times 10^{-2}$ atom %. Field No. is that in Figs. 1 and 2. Polarity from Table 1. Summary of K–Ar age data reported by McDougall and Wensink [3] from lavas in Hnjuksá stream – GA1480, flow 17: 0.253% K, 1.60 ± 0.05 m.y.; GA1479, flow 14: 0.505% K, 2.17 ± 0.10 m.y.

apparent ages are much younger than those determined on sample GL4, which is stratigraphically higher in the sequence. We conclude that there has been appreciable loss of radiogenic argon from GL2. Because of this problem caution must be exercised in the interpretation of all the K–Ar ages: the possibility exists that all the samples may have lost variable amounts of radiogenic argon from the glass present.

Samples GL1 and GH16 were collected from localities about 1.3 km apart (Fig. 1). From their similar petrographies, potassium contents and directions of remanent magnetism (Table 1) it is virtually certain that they are from the same flow, as earlier suspected in the field (Fig. 2). The measured K–Ar ages, indistinguishable at 1.53 and 1.55 m.y. respectively, confirm this point. This unit is the stratigraphically highest normal polarity lava in the sequence, and is very likely the same flow which provided sample GA1480 dated at 1.60 ± 0.05 m.y. by McDougall and Wensink [3], leading to the original definition of the Gilsá event. The underlying flow also has normal polarity: three separate samples labelled GH15, GH15-1, and GH15-2, from this flow have average measured ages of 1.68, 1.63 and 1.57 m.y. respectively, yielding a mean age of 1.63 ± 0.06 m.y. Giving unit weight to the results from each of the two normal polarity flows yields a mean age for the Gilsá event of 1.58 ± 0.08 m.y.

A sample (GH11) with reversed polarity from lower in the sequence in Hnjuksá stream (Fig. 2) yields a reproducible age of 2.00 ± 0.06 m.y. (Table 2) consistent with the stratigraphy. Judging from the similarity in potassium contents and petrographies, this is probably the same flow which provided sample GA1479, previously dated by McDougall and Wensink [3].

3.4. The regional peneplain

Two units within the sequence underlying the Arnorsstadahnjukur Tuff-breccia Volcano have been dated. These results (Table 2) show that the volcano

is only 0.1–0.5 m.y. younger than the immediately underlying Hnjuksá stream section. Thus the volcano cannot be assigned to a substantially younger phase of volcanism, following regional tilting and peneplanation of the underlying sequence. Einarsson's [11] apparently reasonable point that such an erosional history would require at least 1 m.y. must therefore be rejected, and instead the data must be interpreted to show that either the erosional surface is only apparent, *or* that such processes can be completed in Iceland in a relatively brief period.

3.5. Polarity history

Our results confirm in large part the original mapping of Wensink [12,13]. He identified the major normal to reversed polarity boundary of the area as the "N_2/R_1" horizon, or the time when that normal polarity period which preceded the present period of normal polarity changed to the most recent period of reversed polarity. This has since become known as the boundary between the Gauss and Matuyama epochs [1,2], and much study has now led to it being placed at $t = 2.41$ m.y. [20]. The earlier published K–Ar data [3] for Wensink's lava numbers 4, 9, 11 and 14 (Fig. 2) confirmed this point.

As shown in Table 1 (b and c), no lava of normal polarity is detected in the R_1 sequence of Tregagilsá and Krengilsá streams, in conflict with the map shown as Fig. 2 in Wensink [13], from which we have derived part of Fig. 1. Only the Gauss/Matuyama boundary is seen. The original Gilsá normal polarity event was named for lava 17 of Wensink (Fig. 2) in the Hnjuksá sequence, at $t = 1.60 \pm 0.05$ m.y. [3]. We believe that our lavas GH15, 15A and 16, and GL1, equivalent to the "NG" group of Einarsson [11], all represent this period of normal polarity.

We have not detected the possibly reversely magnetized lava (Fig. 2) which led McDougall and Wensink ([3], Fig. 1) to suggest that both the Gilsá and Olduvai events were represented in the Hnjuksá section. Since the unit with the lowest precision parameter (reflecting large scatter in remanent magnetism directions) for the entire section occurs within the sequence of four separate units representing the Gilsá event (Table 1a), it is reasonable, however, to suspect that this body may have provided the obviously inconclusive and unpublished date which led Wensink [12,13] to describe the existence of a "possibly" reversely magnetized lava in the section. Table 3 shows our data for each core from what we believe to be the body involved. A decrease in original intensity of magnetization (J_0) by over 90% occurs in some specimens, as the result of demagnetization in peak fields of 100 Oe.

TABLE 3

Paleomagnetic data for unit number 15, Hnjuksá stream section

N	*H* = 0		*H* = 100			*H* = 200		
	D (deg.)	*I* (deg.)	*D* (deg.)	*I* (deg.)	J/J_0	*D* (deg.)	*I* (deg.)	J/J_0
1	153.7	83.7	181.8	80.5	0.14	106.9	40.9	0.25
2	224.3	82.8	193.1	82.8	0.18	40.3	9.0	0.32
3	168.7	85.1	41.4	3.1	0.33	349.5	36.6	0.27
4	271.6	73.2	183.2	87.9	0.09	304.4	73.1	0.07
5	312.7	83.7	309.9	77.6	0.10	118.7	49.1	0.11
6	320.3	78.7	346.6	82.1	0.01	134.1	84.2	0.01
7	42.7	83.4	66.4	84.4	0.03	223.4	66.5	0.04
8	76.2	79.3	86.7	80.7	0.04	22.0	84.4	0.03
	289.2	88.0	42.3	81.7		56.8	74.6	
K	66.7		7.3			3.9		

N = number of cores. *H* = peak demagnetizing field applied in oersteds (*H* = 0 means no treatment). *D* and *I* = declination and inclination of remanent magnetism; *D* in degrees east of north; *I* = degrees below horizontal. J/J_0 = ratio of intensity of magnetization following given demagnetization treatment, to intensity of magnetization in untreated specimen. Last two lines show mean *D* and *I*, and *K* (see Table 1 for definition) for each demagnetizing treatment.

Adjacent lavas, in contrast, lose less than 40% of J_0 after the same treatment. A lava of low stability is indicated. Since stability and J are often related [18], it is not surprising that the mean J of unit 15 is almost an order of magnitude lower than mean J for other lavas in the section. The conspicuous absence of paleomagnetic data for lava 16 (Fig. 2) in table 1 of Wensink [12] is also indirect evidence of magnetic instability (and possibly low J) in this part of the section. Thus we propose that the most probable explanation for the absence of a second normal polarity lava separated from the lava representing the Gilsá event at the type section, is the presence of a low stability, low intensity of magnetization body, inadvertantly identified as reversed polarity in the original survey [12,13] to therefore split an otherwise single normal polarity into an apparent normal to reversed to normal polarity sequence.

4. Conclusions

We have shown that in the type section for the Gilsá event in the stream Hnjuksá two lavas of normal polarity exist, and that there is no reversed polarity lava between them as previously suggested. Our data indicate that the best estimate for the mean minimum age of the Gilsá event is 1.58 ± 0.08 m.y. We are unable to provide a good estimate for its duration.

The major question to be resolved is whether the Gilsá event is distinct from, and younger than, the Olduvai event. Normal polarity rocks at Olduvai Gorge were first recognized by Grommé and Hay [21], and an age of 1.92 = 0.06 m.y. was reported for one of the basalts [22]. Cox et al. [2] recognized that a short duration of normal polarity had occurred with the "R_1" epoch at about t = 1.9 m.y., and this was named the Olduvai event. Further analyses of the Olduvai Gorge K–Ar data [22] by Grommé and Hay [23] unfortunately only allowed them to state that "the span of time represented by these rocks might be as great as 1.6–1.9 m.y. or as little as 1.70–1.75 m.y.". Thus it is not possible to state unequivocally that the Gilsá and Olduvai events are equivalent, as suggested [23], primarily because of the difficulties encountered in dating the rocks that record the normal polarity events in the respective type sections. Totally satisfactory resolution of this problem requires, therefore, detailed paleomagnetic and dating studies on a thick sequence of fresh subaerial lavas which were erupted during the early Matuyama epoch, which is now known to include a normal polarity event (known as the Reunion event) at t = 2.01 ± 0.01 m.y. [16], older than the Gilsá and the Olduvai events.

Acknowledgements

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Appendix

Petrographic descriptions of dated samples

Lava GL4. Basalt with sporadic plagioclase phenocrysts. Well-crystallized groundmass of plagioclase, clinopyroxene and iron oxide with about 2% of pale brown, slightly anisotropic glass.

Lava GL2. Basalt with about 10% plagioclase and clinopyroxene phenocrysts, and rare fresh olivine microphenocrysts set in a basaltic groundmass containing about 3% pale brown glass.

Lava GH16. Basalt containing about 5% plagioclase phenocrysts, sometimes in aggregates, in a well-crystallized groundmass with about 2% pale brown, isotropic to weakly anisotropic glass. Minor fresh olivine in the groundmass. Somewhat vesicular.

Lava GH15 and specimens GH15-1, GH15-2. Very similar to GH16.

Lava GH12A. Similar to GH16, but the glass shows slight alteration.

Lava GH11. Sporadic plagioclase phenocrysts in slightly fluidal medium-grained, well-crystallized groundmass containing several percent of pale brown, slightly altered, anisotropic glass.

References

1 N.D. Watkins, Review of the development of the geomagnetic polarity time scale and discussion of prospects for its finer definition, Bull. Geol. Soc. Am. 83 (1972) 551–574.

2 A. Cox, R.R. Doell and G.B. Dalrymple, Reversals of the earth's magnetic field, Science 144 (1964) 1537–1543.
3 I. McDougall and H. Wensink, Paleomagnetism and geochronology of the Pliocene-Pleistocene lavas in Iceland, Earth Planet. Sci. Lett. 1 (1966) 232–236.
4 F.J. Vine, Magnetic anomalies associated with mid-oceanic ridges, in: R.A. Phinney, ed., The History of the Earth's Crust (Princeton Univ. Press, Princeton, N.J., 1968) 73–89.
5 J.R. Heirtzler, G.O. Dickson, E.N. Herron, W.C. Pitman and X. LePichon, Marine magnetic anomalies, geomagnetic field reversals and motions of the ocean floor and continents, J. Geophys. Res. 73 (1968) 2119–2136.
6 D.A. Emilia and D.F. Heinrichs, Ocean-floor spreading: Olduvai and Gilsá events in the Matuyama epoch, Science 166 (1969) 1267–1269.
7 N.D. Watkins and H.G. Goodell, Confirmation of the reality of the Gilsá geomagnetic polarity event, Earth Planet. Sci. Lett. 2 (1967) 123–129.
8 D. Ninkovich, N.D. Opdyke, B.C. Heezen and J. Foster, Paleomagnetic stratigraphy, rates of deposition, and tephrachronology in North Pacific deep-sea sediments, Earth Planet. Sci. Lett. 1 (1966) 476–492.
9 N.D. Opdyke and B.P. Glass, The paleomagnetism of sediment cores from the Indian Ocean, Deep Sea Res. 16 (1969) 249–261.
10 N.D. Opdyke, Paleomagnetism of deep-sea cores, Rev. Geophys. Space Phys. 10 (1972) 213–249.
11 Tr. Einarsson, Magnetic polarity groups in Fljotsdalsheidi area, including Gilsá, Jokull 21 (1972) 53–58.
12 H. Wensink, Secular variation of earth magnetism in Plio-Pleistocene basalts of eastern Iceland, Geol. Mijnb. 43 (1964) 403–413.
13 H. Wensink, Paleomagnetic stratigraphy of younger basalts and intercalated Plio-Pleistocene tillites in Iceland, Geol. Rundsch. 54 (1964) 364–384.
14 N.D. Watkins, A. Hajash and C.E. Abranson, Geomagnetic secular variation during the Brunhes epoch in the Indian and Atlantic Ocean regions, Geophys. J.R. Astron. Soc. 28 (1972) 1–25.
15 J.A. Cooper, The flame photometric determination of potassium in geological materials used in potassium-argon dating, Geochim. Cosmochim. Acta 27 (1963) 525–546.
16 I. McDougall and N.D. Watkins, Age and duration of the Reunion geomagnetic polarity event, Earth Planet. Sci. Lett. 19 (1973) 443–452.
17 R. Fisher, Dispersion on a sphere, Proc. R. Soc. London A217 (1953) 295–305.
18 N.D. Watkins and S.E. Haggerty, Primary oxidation variation and petrogenesis in a single lava, Contrib. Mineral. Petrol. 15 (1967) 251–271.
19 I. McDougall, H.A. Polach, and J.J. Stipp, Excess radiogenic argon in young subaerial basalts from the Auckland volcanic field, New Zealand, Geochim. Cosmochim. Acta 33 (1969) 1485–1520.
20 I. McDougall and Aziz-Ur-Rahman, Age of the Gauss-Matuyama boundary and of the Kaena and Mammoth events, Earth Planet. Sci. Lett. 14 (1972) 367–380.
21 C.S. Grommé and R.L. Hay, Magnetization of basalt of Bed 1, Olduvai Gorge, Tanganyika, Nature 200 (1963) 560–561.
22 J.F. Evernden and G.H. Curtis, The potassium-argon dating of late Cenozoic rocks in East Africa and Italy, Current Anthropol. 6 (1965) 343–385.
23 C.S. Grommé and R.L. Hay, Geomagnetic polarity epochs: age and duration of the Olduvai normal polarity event, Earth Planet. Sci. Lett. 10 (1971) 179–185.